

Git Fork, Merge, Conflict & Rebase - Complete Steps

Step 1: Clone Repository

Command:

```
git clone https://github.com/reddy/shopizer.git
```

Next:

```
cd shopizer
```

Step 2: Create Feature Branch

Command:

```
git checkout -b feature-branch
```

Next:

```
echo "Hello DevOps Lab" > file.txt
```

```
git add file.txt
```

```
git status
```

```
git commit -m "Added file.txt"
```

Step 3: Modify File

Command:

```
echo "Welcome to DevOps Lab" >> file.txt
```

Next:

```
git add file.txt
```

```
git commit -m "Added new line"
```

Step 4: Push to GitHub

Commands:

```
git remote add origin https://github.com//.git
```

```
git remote -v
```

```
git push origin feature-branch
```

Next:

Create Pull Request and Merge on GitHub

Step 5: Merge Branch

Commands:

```
git checkout main
```

```
git merge feature-branch
```

Step 6: Create Branch1

Commands:

```
git checkout -b branch1
echo "Line from branch1" > file.txt
git add file.txt
git commit -m "branch1 change"
```

Next:

```
git checkout main
```

Step 7: Create Branch2

Commands:

```
git checkout -b branch2
echo "Line from branch2" > file.txt
git add file.txt
git commit -m "branch2 change"
```

Next:

```
git checkout branch1
```

Step 8: Create Conflict

Command:

```
git merge branch2
```

Next:

Open file.txt and resolve conflict

Step 9: Resolve Conflict

Edit file.txt and keep both lines.

Commands:

```
git add file.txt
git commit -m "Resolved conflict"
```

Step 10: Main Update

Commands:

```
git checkout main
echo "Main Update" >> file.txt
git add file.txt
git commit -m "main update"
```

Step 11: Rebase

Commands:

```
git checkout feature-branch
git rebase main
```

Result:

Successfully rebased and updated refs/heads/feature-branch